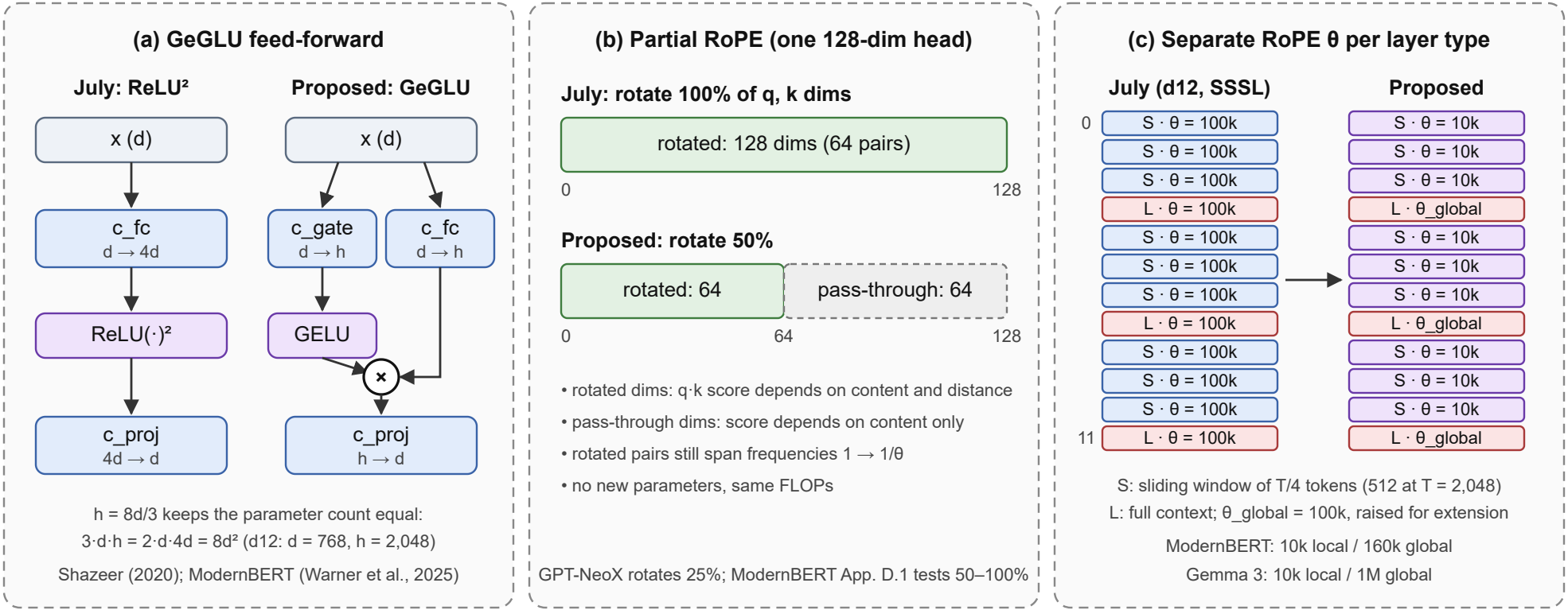

(a) GeGLU feed-forward
July: ReLU²
Proposed: GeGLU
x (d)
c_fc
d → 4d
ReLU(·)²
c_proj
4d → d
x (d)
c_gate
d → h
c_fc
d → h
GELU
×
c_proj
h → d
h = 8d/3 keeps the parameter count equal:
3·d·h = 2·d·4d = 8d² (d12: d = 768, h = 2,048)
Shazeer (2020); ModernBERT (Warner et al., 2025)
(b) Partial RoPE (one 128-dim head)
July: rotate 100% of q, k dims
rotated: 128 dims (64 pairs)
0
128
Proposed: rotate 50%
rotated: 64
pass-through: 64
0
64
128
• rotated dims: q·k score depends on content and distance
• pass-through dims: score depends on content only
• rotated pairs still span frequencies 1 → 1/θ
• no new parameters, same FLOPs
GPT-NeoX rotates 25%; ModernBERT App. D.1 tests 50–100%
(c) Separate RoPE θ per layer type
July (d12, SSSL)
Proposed
0
S · θ = 100k
S · θ = 100k
S · θ = 100k
L · θ = 100k
S · θ = 100k
S · θ = 100k
S · θ = 100k
L · θ = 100k
S · θ = 100k
S · θ = 100k
S · θ = 100k
11
L · θ = 100k
S · θ = 10k
S · θ = 10k
S · θ = 10k
L · θ_global
S · θ = 10k
S · θ = 10k
S · θ = 10k
L · θ_global
S · θ = 10k
S · θ = 10k
S · θ = 10k
L · θ_global
S: sliding window of T/4 tokens (512 at T = 2,048)
L: full context; θ_global = 100k, raised for extension
ModernBERT: 10k local / 160k global
Gemma 3: 10k local / 1M global